

## Unit 115 - Field Data Review

Mohawk Valley West Slope demonstration. Method **Tractor**, 84.0 ac, BAF 20 variable-radius cruise on a 300 ft grid, 39 plots, 444 trees tallied. Simulated data with planted recording errors; see docs/methods.md.

**Table 1. Stand summary from the BAF 20 cruise, per acre**

| STAND DENSITY        |                                    | STAND INFO          |                                                |
|----------------------|------------------------------------|---------------------|------------------------------------------------|
| Basal area           | 228 sq ft/ac (SE 2.0 %)            | Type                | Sierran mixed conifer                          |
| Trees per acre       | 87                                 | CWHR size / density | 4 / D (cover 92 %)                             |
| QMD                  | 21.9 in                            | SDI                 | 292 (51 % of max 578)                          |
| SAWTIMBER (>= 10 in) |                                    | BIOMASS (< 10 in)   |                                                |
| BA   QMD             | 227 sq ft/ac   22.2 in             | BA   QMD            | 1 sq ft/ac   8.2 in                            |
| TPA                  | 85                                 | TPA                 | 3                                              |
| Volume               | 47.5 MBF/ac (8636 cu ft)           | Volume              | 0.6 green tons/ac (21 cu ft)                   |
| TREATMENT TARGET     |                                    | CRUISE DESIGN       |                                                |
| Leave                | 35 % of max SDI = ~158 sq ft/ac BA | Sampling error      | 4.0 % at 95 % (t = 2.02, n = 39)               |
| Species (BA)         | WF 68, PP 67, IC 34, DF 33, SP 26  | Stratum standard    | 40 % - MEETS; 39 plots vs 20-plot practice: ok |

Cubic volume by form factor 0.42 net of defect; Scribner board feet at 5.5 bf/cu ft (Keegan et al. 2010); green tons at 60 lb/cu ft. SDI in the summation form; maximum is the basal-area-weighted FVS Western Sierra species maximum. Sampling error at 95 % confidence with Student's t; stratum standard 40 % for tree-measurement sales (FSH 2409.12 ch. 40, 41.1).

**Table 2. Stand table by 4-inch DBH class, per acre**

| DBH class, in | Trees      | TPA         | BA           | cu ft       | MBF         |
|---------------|------------|-------------|--------------|-------------|-------------|
| 4-7.9         | 1          | 1.8         | 0.5          | 10          | 0.1         |
| 8-11.9        | 11         | 8.8         | 5.6          | 133         | 0.7         |
| 12-15.9       | 43         | 20.2        | 22.1         | 615         | 3.4         |
| 16-19.9       | 64         | 18.7        | 32.8         | 971         | 5.3         |
| 20-23.9       | 73         | 14.2        | 37.4         | 1308        | 7.2         |
| 24-27.9       | 68         | 9.8         | 34.9         | 1342        | 7.4         |
| 28-31.9       | 51         | 5.4         | 26.2         | 1084        | 6.0         |
| 32-35.9       | 48         | 3.9         | 24.6         | 1083        | 6.0         |
| 36-39.9       | 31         | 2.1         | 15.9         | 731         | 4.0         |
| 40-43.9       | 22         | 1.2         | 11.3         | 521         | 2.9         |
| 44-47.9       | 15         | 0.7         | 7.7          | 381         | 2.1         |
| 48-51.9       | 9          | 0.3         | 4.6          | 249         | 1.4         |
| 52-55.9       | 5          | 0.2         | 2.6          | 132         | 0.7         |
| 60-63.9       | 3          | 0.1         | 1.5          | 97          | 0.5         |
| <b>All</b>    | <b>444</b> | <b>87.3</b> | <b>227.7</b> | <b>8657</b> | <b>47.6</b> |

**Table 3. Stock table by species, per acre**

| Species    | Trees      | TPA         | BA           | QMD         | cu ft       | MBF         |
|------------|------------|-------------|--------------|-------------|-------------|-------------|
| WF         | 132        | 24.4        | 67.7         | 22.6        | 2613        | 14.4        |
| PP         | 130        | 24.9        | 66.7         | 22.2        | 2557        | 14.1        |
| IC         | 66         | 14.5        | 33.8         | 20.7        | 1223        | 6.7         |
| DF         | 65         | 12.4        | 33.3         | 22.2        | 1317        | 7.2         |
| SP         | 51         | 11.2        | 26.2         | 20.7        | 948         | 5.2         |
| <b>All</b> | <b>444</b> | <b>87.3</b> | <b>227.7</b> | <b>21.9</b> | <b>8657</b> | <b>47.6</b> |

Trees = trees tallied in the cruise; TPA, BA (sq ft/ac), cubic feet and Scribner MBF per acre by BAF expansion. Species: PP ponderosa pine, WF white fir, DF Douglas-fir, SP sugar pine, IC incense-cedar.

Unit 115: plots on the 300 ft grid (orange = QA flag)

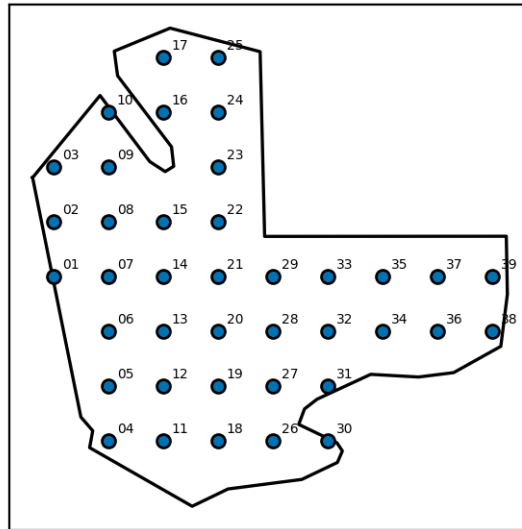


Figure 1. Plot locations with QA status.

**QA findings:** no issues found; unit accepted

**Table 4. Plot list**

| <b>Plot</b> | <b>Trees in</b> | <b>BA sq ft/ac</b> | <b>Slope %</b> | <b>Status</b> |
|-------------|-----------------|--------------------|----------------|---------------|
| 115-01      | 13              | 260                | 84             | ok            |
| 115-02      | 10              | 200                | 36             | ok            |
| 115-03      | 8               | 160                | 36             | ok            |
| 115-04      | 12              | 240                | 43             | ok            |
| 115-05      | 12              | 240                | 101            | ok            |
| 115-06      | 12              | 240                | 36             | ok            |
| 115-07      | 11              | 220                | 144            | ok            |
| 115-08      | 11              | 220                | 38             | ok            |
| 115-09      | 11              | 220                | 43             | ok            |
| 115-10      | 9               | 180                | 91             | ok            |
| 115-11      | 12              | 240                | 24             | ok            |
| 115-12      | 12              | 240                | 29             | ok            |
| 115-13      | 10              | 200                | 52             | ok            |
| 115-14      | 10              | 200                | 30             | ok            |
| 115-15      | 12              | 240                | 29             | ok            |
| 115-16      | 9               | 180                | 104            | ok            |
| 115-17      | 11              | 220                | 96             | ok            |
| 115-18      | 11              | 220                | 25             | ok            |
| 115-19      | 8               | 160                | 24             | ok            |
| 115-20      | 11              | 220                | 38             | ok            |
| 115-21      | 11              | 220                | 81             | ok            |
| 115-22      | 14              | 280                | 41             | ok            |
| 115-23      | 12              | 240                | 22             | ok            |
| 115-24      | 12              | 240                | 53             | ok            |
| 115-25      | 12              | 240                | 65             | ok            |
| 115-26      | 10              | 200                | 24             | ok            |
| 115-27      | 12              | 240                | 20             | ok            |
| 115-28      | 12              | 240                | 38             | ok            |
| 115-29      | 12              | 240                | 16             | ok            |
| 115-30      | 11              | 220                | 11             | ok            |
| 115-31      | 12              | 240                | 11             | ok            |
| 115-32      | 12              | 240                | 36             | ok            |
| 115-33      | 12              | 240                | 11             | ok            |
| 115-34      | 12              | 240                | 37             | ok            |
| 115-35      | 13              | 260                | 12             | ok            |
| 115-36      | 15              | 300                | 14             | ok            |
| 115-37      | 11              | 220                | 37             | ok            |
| 115-38      | 12              | 240                | 21             | ok            |
| 115-39      | 12              | 240                | 25             | ok            |